Plasma and erythrocyte biomarkers of dairy fat intake and risk of ischemic heart disease.

BACKGROUND: The relation between dairy product intake and the risk of ischemic heart disease (IHD) remains controversial. OBJECTIVE: We aimed to explore biomarkers of dairy fat intake in plasma and erythrocytes and to assess the hypothesis that higher concentrations of these biomarkers are associated with a greater risk of IHD in US women. DESIGN: Among 32,826 participants in the Nurses' Health Study who provided blood samples in 1989-1990, 166 incident cases of IHD were ascertained between baseline and 1996. These cases were matched with 327 controls for age, smoking, fasting status, and date of blood drawing. RESULTS: Among controls, correlation coefficients between average dairy fat intake in 1986-1990 and 15:0 and trans 16:1n-7 content were 0.36 and 0.30 for plasma and 0.30 and 0.32 for erythrocytes, respectively. In multivariate analyses, with control for age, smoking, and other risk factors of IHD, women with higher plasma concentrations of 15:0 had a significantly higher risk of IHD. The multivariate-adjusted relative risks (95% CI) from the lowest to highest tertile of 15:0 concentrations in plasma were 1.0 (reference), 2.18 (1.20, 3.98), and 2.36 (1.16, 4.78) (P for trend = 0.03). Associations for other biomarkers were not significant. CONCLUSIONS: Plasma and erythrocyte contents of 15:0 and trans 16:1n-7 can be used as biomarkers of dairy fat intake. These data suggest that a high intake of dairy fat is associated with a greater risk of IHD.